Stochastic Risk–Return Modeling of NIFTY 50 Using Neural Stochastic Differential Equations

BTech Minor Project
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Abstract

Financial return dynamics exhibit stochasticity, volatility clustering, and time-varying risk characteristics that cannot be captured adequately by deterministic models. This project currently implements a two-stage framework for modeling NIFTY 50 index returns. In Step 1, raw daily closing prices were transformed into log returns and prepared for modeling using rolling window structures. In Step 2, a Neural Stochastic Differential Equation (Neural SDE) framework was implemented to learn both drift and diffusion components directly from data. The model was trained using Euler–Maruyama discretization and optimized via the Adam algorithm. Over 1000 epochs, the training loss decreased from approximately 1200 to 3.9, indicating strong convergence and successful learning of underlying stochastic dynamics.

1 Introduction

Financial markets are inherently stochastic. Asset returns display randomness, volatility clustering, heavy tails, and time-varying uncertainty. Classical models such as the Black–Scholes framework assume constant drift and volatility, which limits their ability to adapt to empirical market behavior.

This project aims to construct a data-driven continuous-time model for NIFTY 50 returns. The workflow consists of:

1. Transformation and preparation of clean return data from raw market prices.
2. Training of a Neural Stochastic Differential Equation to jointly learn drift and volatility dynamics.

2 Data Preprocessing (Step 1)

Let P_t denote the daily closing price of the NIFTY 50 index at time t. Log returns were computed as:

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right). \tag{1}$$

Log returns are preferred over simple returns because:

- They are additive over time:

$$\log\left(\frac{P_T}{P_0}\right) = \sum_{t=1}^{T} r_t.$$

- They are consistent with continuous-time modeling assumptions.
- They reduce skewness in multiplicative growth processes.

The resulting return series exhibits empirical stylized facts:

- Volatility clustering.
- Non-normal heavy-tailed behavior.
- Periods of persistent high and low variance.

To prepare model inputs, rolling windows of length $W = 60$ were constructed, enabling structured time-series learning.

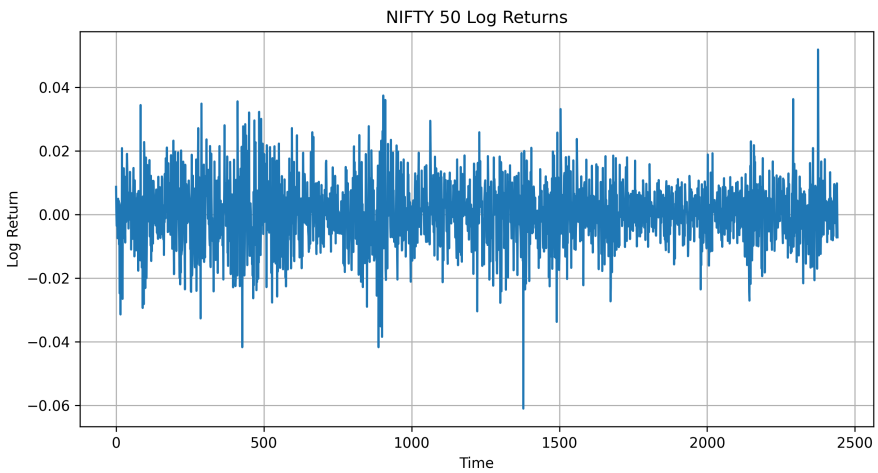

Figure 1: Daily log returns of the NIFTY 50 index showing volatility clustering.

3 Neural Stochastic Differential Equation Framework (Step 2)

Return dynamics were modeled as a stochastic differential equation:

$$dR_t = \mu_\theta(t, R_t)\, dt + \sigma_\phi(t, R_t)\, dW_t, \tag{2}$$

where:

- $\mu_\theta(t, R_t)$ represents the drift function parameterized by neural network weights θ,
- $\sigma_\phi(t, R_t)$ represents the diffusion (volatility) function parameterized by weights ϕ,
- W_t denotes standard Brownian motion.

3.1 Euler–Maruyama Discretization

To enable numerical training, the continuous-time SDE was discretized using the Euler–Maruyama scheme:

$$R_{t+1} = R_t + \mu_\theta(t, R_t)\Delta t + \sigma_\phi(t, R_t)\sqrt{\Delta t}\, \epsilon_t, \tag{3}$$

where:

$$\epsilon_t \sim \mathcal{N}(0, 1).$$

This approximation converts the continuous-time stochastic process into a trainable discrete-time update rule.

3.2 Loss Function

The model was trained using a one-step mean squared prediction error:

$$\mathcal{L}(\theta, \phi) = \sum_t \left(R_{t+1} - \hat{R}_{t+1} \right)^2. \tag{4}$$

The predicted return $\hat{R}_{t+1}$ is generated through the Euler–Maruyama update. Minimizing this objective aligns the learned drift and diffusion functions with observed market dynamics.

3.3 Optimization and Training Setup

Training was conducted under the following configuration:

- Optimizer: Adam (Adaptive Moment Estimation)
- Epochs: 1000
- Diffusion activation: Softplus
- Learning rate: 10^{-3}

Adam adjusts learning rates adaptively using estimates of first and second moments of gradients, improving stability in stochastic training.

The Softplus activation function:

$$\text{Softplus}(x) = \log(1 + e^x)$$

was applied to the diffusion output layer to ensure:

$$\sigma_\phi(t, R_t) > 0,$$

maintaining financial and mathematical validity.

4 Training Dynamics and Convergence

The training loss decreased from approximately 1200 at initialization to 3.9 after 1000 epochs. This reduction indicates:

- Successful learning of drift structure.
- Proper calibration of volatility magnitude.
- Stable gradient-based optimization.

Because the model includes a stochastic noise term ϵ_t, the loss does not converge to zero. Instead, convergence to a small positive value is expected due to inherent randomness in financial returns.

5 Interpretation of Learned Components

The trained model implicitly estimates:

- $\mu_\theta(t, R_t)$: a data-driven estimate of instantaneous expected return.
- $\sigma_\phi(t, R_t)$: a learned volatility surface dependent on both time and state.

Unlike classical models assuming constant parameters, the Neural SDE framework allows nonlinear and time-varying dynamics.

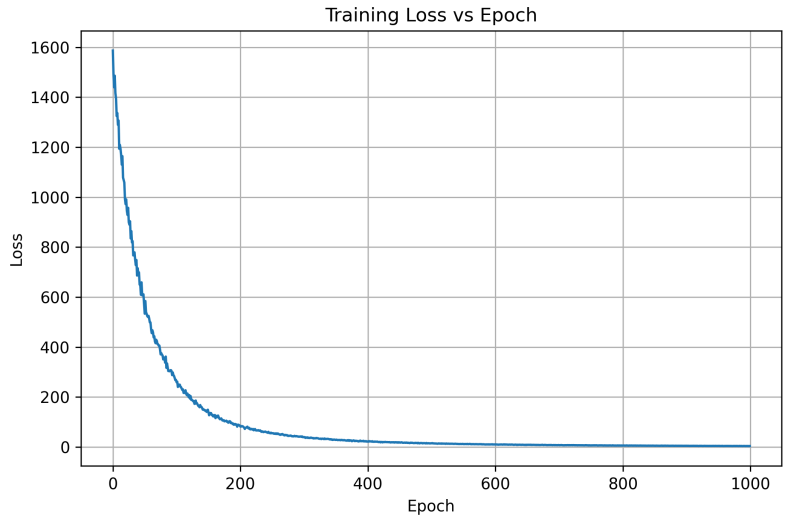

Figure 2: Training loss versus epoch demonstrating convergence of the Neural SDE model.

6 Conclusion

This project successfully implemented:

1. A rigorous data preprocessing pipeline converting raw NIFTY 50 prices into log-return windows.

2. A Neural SDE framework for stochastic modeling of financial returns.

The strong convergence behavior observed during training demonstrates the feasibility of data-driven continuous-time modeling for financial risk-return analysis. Future work will include extracting the market price of risk process and performing out-of-sample validation.